Transporting the Cross-Modal Prototypes for Unsupervised Visible-Infrared Person Re-Identification

Zhizhong Zhang, Jiaqi Chen, Xin Tan, *Member, IEEE*, Yuan Xie, *Member, IEEE*, and Yan Gao

***Abstract*—Unsupervised visible infrared person re-identification (USVI-ReID) is a challenging retrieval task that retrieves cross-modality pedestrian images without using any label information. In this task, the large cross-modality variance makes it difficult to generate reliable cross-modality labels, and the lack of annotations also provides additional difficulties for learning modality-invariant features. To facilitate this unsupervised cross-modal learning, we begin by leveraging the information contained in the cross-modality input and its predicted label. Aiming to minimize information loss, we optimize the model by incorporating entropy minimization, uniform label distribution, and cross-modality matching. In our approach, we design a loop iterative training strategy alternating between model training and cross-modality matching, where a uniform prior guided optimal transport assignment is proposed to select matched visible and infrared prototypes. This matching information is then utilized to minimize the intra- and cross-modality entropy. As a result, our model can gradually self-learn useful information, enabling it to generate discriminative representations for unlabeled cross-modal data. Extensive experimental results on benchmarks demonstrate the effectiveness of our method, *e.g.*, 69.4% and 89.4% of Rank-1 accuracy on SYSU-MM01 and RegDB without any annotations. The code will be released soon.**

***Index Terms*—Unsupervised learning, visible infrared person re-identification, optimal transport, mutual information.**

I. Introduction

THE purpose of person re-identification (ReID) is to match the pedestrian images across multiple non-overlapping camera views. Most ReID methods [1], [2], [3], [4], [5] focus on the daytime setting due to the easy acquisition of colorful visible images. But with the development of sensor technology, *e.g.*, infrared cameras, clear images can also be captured under poor illumination conditions. Therefore, visible-infrared person re-identification (VI-ReID) emerges as a critical enabler for real-world security and intelligent transportation systems. These applications demand continuous 24-hour monitoring capabilities, which allows us to retrieve the target images (visible or infrared) when provided with a query image from a different modality [6], [7], [8], [9], [10], [11], [12].

However, due to the substantial differences in sensing processes, these heterogeneous images exhibit considerable variations in appearance and color. For example, infrared imaging relies on the detection of thermal radiation, thereby emphasizing shape, contour, and temperature distribution while lacking detailed visual attributes such as color and fine-grained texture [13]. Thus features that are discriminative in visible images—such as clothing color—may be entirely absent or misrepresented in their infrared counterparts, thereby increasing the complexity of associating corresponding pedestrian instances, which significantly complicates the manual labeling process [14].

To deal with this issue, a growing body of works [16], [17], [18], [19], [20] has focused on training a discriminative VI-ReID model without infrared label information or complete annotation. A common practice is to generate the cluster-aware pseudo labels, which are then utilized to supervise the training of model [21], [22]. However, the significant cross-modality discrepancy leads to difficulties in generating reliable cross-modality pseudo labels, while the lack of annotations also provides additional difficulties in learning discriminative modality-invariant features. The compounding effect renders unsupervised cross-modal training particularly difficult, requiring the simultaneous learning of cross-modal correspondences and identity information from the images themselves.

In this paper, we begin by leveraging and measuring the information contained in the cross-modality input and its predicted label. As shown in Fig. 1, we extend the classic unsupervised framework ReMixMatch [15] to cross-modal unsupervised representation, by compressing the cross-modality input with minimal information losing. It is then transferred to three learning priors: entropy minimization, uniform label distribution and cross-modality matching. These priors collectively enhance the model's ability to learn

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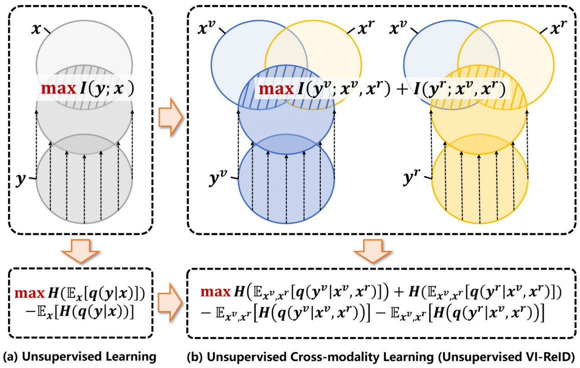

Fig. 1. (a) One way to realize the unsupervised learning is to maximize the mutual information between the model's input $\boldsymbol{x}$ and output $\boldsymbol{y}$ [15]. (b) For unsupervised VI-ReID task, we additionally mine the modality-consistent mutual information (*i.e.*, maximize the mutual information between the matched cross-modality input pairs $\boldsymbol{x}^v, \boldsymbol{x}^r$ and the corresponding outputs $\boldsymbol{y}^v, \boldsymbol{y}^r$, where v and r denote visible and infrared modality respectively).

robust and generalizable representations under this inherently entangled task.

In our implementation, we design a loop iterative training strategy alternating between model training and cross-modality matching. Prior to each training epoch, DBSCAN [23] is employed to generate coarse pseudo labels from each modal, enabling to obtain the cluster-aware and modal-specific prototypes. Then to match the cross-modal data, we propose a dual transport framework, where a both forward and backward transport is adopted to merge the cross-modality prototypes with higher transport probability. In this matching stage, a uniform label distribution is introduced in this transporting to overcome all prototypes being grouped into one. Finally, we optimize our model by the contrastive learning through the correspondence information, an equivalent substitute for entropy minimization. Alternating between cross-modality matching and learning, our model can gradually self-learn useful information to produce discriminative representation for unlabeled cross-modal data. Empirical results show the strong robustness of our transporting framework. Even when the identities from visible and infrared images are not completely overlapping, the learned correspondence still enables effective optimization of our model. In general, our contributions can be summarized as follows:

- We decouple the learning objective of unsupervised cross-modal training, by maximizing the mutual information between cross-modal input and output. It therefore allows for optimizing the model in terms of entropy minimization, uniform label distribution and cross-modality matching.
- We formulate the cross-modality prototype matching problem as a dual optimal transport problem, where both forward and backward transportation assignments ensure that cross-modal prototypes are effectively matched. Based on this matching information, entropy is then well minimized.
- We conduct extensive experiments on widely adopted VI-ReID benchmarks. Empirical results show that our proposed method achieves great improvement against a series of state-of-the-art unsupervised learning ReID (VI-ReID) methods, also highly comparable against fully supervised VI-ReID methods.

II. Related Work

A. Person Re-Identification

1) Unsupervised Learning Person ReID (USL-ReID): Unsupervised person re-identification is a single-modality ReID task, which aims to train a model with only unlabeled visible pedestrian data. In this setting, nearly all of the methods [4], [5], [24], [25], [26], [27], [28], [29] are pseudo-labels-based, which then establish a bridge towards the supervised manner via these pseudo labels. For example, several works [5], [25], [26], [27], [28] try to obtain pseudo labels with the traditional clustering method(*e.g.*, DBSCAN or K-means) and then gradually improve the quality of pseudo labels by refining clusters as the training goes on. Specifically, ISE [3] uses the progressive linear interpolation of KNN samples to update the clusters. While [5], [25], [26] gets the help of contrastive learning loss with the special design of clustering memory bank. As for PAT [28], it benefits from a mutual learning mechanism through a shared Transformer architecture. Besides, some hierarchical clustering ways [24], [30], [31] with merge-split cluster's refinement are proposed to obtain high-quality pseudo labels from another perspective.

2) Supervised Visible-Infrared Person ReID (SVI-ReID): It is a challenging cross-modality image retrieval problem, which aims to match the pedestrian images captured by visible and infrared cameras. Up to now, impressive progress has been made in it. To this end, most works [32], [33], [34], [35], [36], [37], [38], [39], [40] in this community try to enhance the feature discrimination by introducing novel network architectures (*e.g.*, graph convolution network, non-local module, transformer or self-designed modality fusion model). Some of these works [34], [41], [42], [43], [44] also can discover nuanced but discriminative clues (*e.g.*, part or patch-level information) for both modalities so as to improve the quality of feature representation. While, another line of works [38], [39], [45], [46], [47], [48], [49] attempts to excavate modality-invariant information via cross-modality image generation or mix-up manipulation. Besides, [36], [50], [51], [52] have optimized metric learning losses (*e.g.*, triplet loss or memory-based contrastive learning loss) adapting to cross-modality learning.

3) Unsupervised Visible-Infrared Person ReID (USVI-ReID): Recently, this task has gradually drawn great attention due to the laboring efforts of annotation, especially for infrared data without color information. Some works [14], [16] try to solve this problem with the assistance of other labeled visible datasets (*e.g.*, Market-1501 [53]). Specifically, OTLA-ReID [14] utilizes the standard unsupervised domain adaptation technique to generate the pseudo labels for the visible subset with the help of well-annotated visible datasets, and then leverage optimal transport theory to facilitate cross-modality learning. Other works [17], [19], [20], [22], [54], [55], [56], [57] directly apply self-supervised learning on unlabeled cross-modality datasets. Among them, some work [22], [54], [55], [56], [58] employ bipartite graph matching to uncover cross-modality correspondence. Besides, PCLHD [59] leverages hard and dynamic prototypes in progressive contrastive learning to capture commonality, divergence, and variety.

B. Optimal Transport

Optimal transport (OT) is often used to find connections between variables or measure the probability distribution distance, which already has obtained increasing attention in machine learning. One of the widely used implementations is the Sinkhorn-Knopp algorithm [60], which decreases the computation cost of OT problems significantly. SeLa [61] has extended OT to self-supervised learning. In this framework, they alternate between the following two steps: 1) Making use of the Sinkhorn-Knopp algorithm to produce pseudo labels for unlabeled data. 2) Doing classification with current pseudo labels. In fact, SeLa [61] is a clustering-based self-supervised approach, which aims to find a good pretraining model. Recent advances [14], [19], [62], [63] have increasingly focused on applying OT to tackle USVI-ReID challenges. The most related work to ours is OTLA-ReID [14]. However, it mainly focuses on the semi-supervised task and applies OT for a sample-label assignment problem with the help of the classifier. Obviously, it's much harder to define the proper classifier without any real label information. Thus it doesn't work well for the unsupervised task. By contrast, our proposed method mainly focuses on the unsupervised task and applies OT for a cross-modality prototype assignment problem without using a classifier, which also achieves better performance than OTLA-ReID.

III. Mutual Information Guided Prior

In this section, we first briefly review the mutual information [64] and its application ReMixMatch [15]. ReMixMatch offers a well-defined learning objective for unsupervised learning, and we extend ReMixMatch to cross-modal representation by comparing the information between cross-modality inputs and their predicted labels. Our results demonstrate that such cross-modal model without annotation can be effectively optimized in terms of entropy minimization, label distribution uniformity, and cross-modality matching.

A. Preliminary

Given the input data $\boldsymbol{x}$, some works [65], [66], [67], [68] start from the information bottleneck (IB) by measuring the mutual information between $\boldsymbol{x}$ and its compressed representation z. Actually, the objective of IB *i.e.,* $\max I(\boldsymbol{x}; z)$, is a principle of feature compression, which makes the representation z contain as much useful information from the input data $\boldsymbol{x}$ as possible. Furthermore, ReMixMatch [15] and related works [15], [69], [70], [71] directly maximize the mutual information between the model's input $\boldsymbol{x}$ and output $\boldsymbol{y}$ (*i.e.*, predicted label in unsupervised learning):

$$\max \; I(\boldsymbol{y}; \boldsymbol{x}) = \iint p(\boldsymbol{y}, \boldsymbol{x}) \log\left(\frac{p(\boldsymbol{y}, \boldsymbol{x})}{p(\boldsymbol{y})p(\boldsymbol{x})}\right) d\boldsymbol{y} d\boldsymbol{x}. \tag{1}$$

According to variational inference, we can obtain the variational lower bound of $I(\boldsymbol{x}; \boldsymbol{y})$ as a combination consisting of a negative entropy term augmented with a regularization term:

$$I(\boldsymbol{y}; \boldsymbol{x}) \geq \underbrace{H(\mathbb{E}_{\boldsymbol{x}}[q(\boldsymbol{y}|\boldsymbol{x})])}_{\text{Regularization}} - \underbrace{\mathbb{E}_{\boldsymbol{x}}[H(q(\boldsymbol{y}|\boldsymbol{x}))]}_{\text{Entropy}}, \tag{2}$$

where $q(\cdot)$ is an arbitrary variational distribution, $H(\cdot)$ denotes the entropy, and $\mathbb{E}[\cdot]$ denotes the expectation value. Note that Eq. (2) is a semi-supervised objective from ReMixMatch, by using variational distribution $q(\boldsymbol{y}|\boldsymbol{x})$ to replace $p(\boldsymbol{y}|\boldsymbol{x})$ for approximating its lower bound.

1) Estimation of $q(\cdot)$: For the semi-supervised learning setting, such as ReMixMatch, $q(\cdot)$ can be transferred into the classifier's prediction with softmax manipulation. However, we cannot define a proper parameterized classifier under the unsupervised setting. Here we use clustering methods to obtain prototypes (*e.g.*, mean of latent feature representations) to imitate the column parameter vectors of the classifier. Then we transfer $q(\cdot)$ into a prototype-based cosine similarity vector with softmax manipulation, inspired by few-shot learning [72].

B. Unsupervised Cross-Modal Representation Learning

We therefore modify the learning objective of Eq. (1) to extend it for unsupervised cross-modal representation:

$$\max \; I(\boldsymbol{y}^v; \boldsymbol{x}^v) + I(\boldsymbol{y}^v; \boldsymbol{x}^r) + I(\boldsymbol{y}^r; \boldsymbol{x}^r) + I(\boldsymbol{y}^r; \boldsymbol{x}^v), \tag{3}$$

where $\boldsymbol{x}^v, \boldsymbol{x}^r$ denote the hypothetically matched cross-modality input pair, and $\boldsymbol{y}^v, \boldsymbol{y}^r$ denote corresponding model's predicted label, *e.g.*, pseudo label. Eq. (3) implies that the model's output should contain the essential information for both intra- and cross-modality input of the same identity. Maximizing Eq. (3) hence offers a clear objective for cross-modal unsupervised learning, which, furthermore, can be transferred to:

$$\max \; I(\boldsymbol{y}^v; \boldsymbol{x}^v, \boldsymbol{x}^r) + I(\boldsymbol{y}^r; \boldsymbol{x}^v, \boldsymbol{x}^r), \tag{4}$$

where $I(\boldsymbol{x}^v, \boldsymbol{x}^r; \boldsymbol{y}^v)$ or $I(\boldsymbol{x}^v, \boldsymbol{x}^r; \boldsymbol{y}^r)$ represents that the information contained in $\boldsymbol{y}^v$ or $\boldsymbol{y}^r$ and shared with the union of $\boldsymbol{x}^v$ and $\boldsymbol{x}^r$. This learning objective additionally mines the modality-consistent mutual information compared with pure unsupervised learning, and hence Eq. (4) simultaneously facilitates unsupervised and cross-modal learning.

Another advantage of Eq. (4) is easy to optimize. In fact, we have its lower bound:

$$\begin{aligned} I(\boldsymbol{y}^v; \boldsymbol{x}^v, \boldsymbol{x}^r) \geq\; & H(\mathbb{E}_{\boldsymbol{x}^v,\boldsymbol{x}^r}[q(\boldsymbol{y}^v|\boldsymbol{x}^v, \boldsymbol{x}^r)]) \\ & - \mathbb{E}_{\boldsymbol{x}^v,\boldsymbol{x}^r}[H(q(\boldsymbol{y}^v|\boldsymbol{x}^v, \boldsymbol{x}^r))], \end{aligned} \tag{5}$$

$$\begin{aligned} I(\boldsymbol{y}^r; \boldsymbol{x}^v, \boldsymbol{x}^r) \geq\; & H(\mathbb{E}_{\boldsymbol{x}^v,\boldsymbol{x}^r}[q(\boldsymbol{y}^r|\boldsymbol{x}^v, \boldsymbol{x}^r)]) \\ & - \mathbb{E}_{\boldsymbol{x}^v,\boldsymbol{x}^r}[H(q(\boldsymbol{y}^r|\boldsymbol{x}^v, \boldsymbol{x}^r))]. \end{aligned} \tag{6}$$

Finally, we obtain the final training objective by combining Eq. (5) and Eq. (6):

$$\begin{aligned} \max \;\; & H(\mathbb{E}_{\boldsymbol{x}^v,\boldsymbol{x}^r}[q(\boldsymbol{y}^v|\boldsymbol{x}^v, \boldsymbol{x}^r)]) \\ & + (\mathbb{E}_{\boldsymbol{x}^v,\boldsymbol{x}^r}[q(\boldsymbol{y}^r|\boldsymbol{x}^v, \boldsymbol{x}^r)]) \\ & - \mathbb{E}_{\boldsymbol{x}^v,\boldsymbol{x}^r}[H(q(\boldsymbol{y}^v|\boldsymbol{x}^v, \boldsymbol{x}^r))] \\ & - \mathbb{E}_{\boldsymbol{x}^v,\boldsymbol{x}^r}[H(q(\boldsymbol{y}^r|\boldsymbol{x}^v, \boldsymbol{x}^r))]. \end{aligned} \tag{7}$$

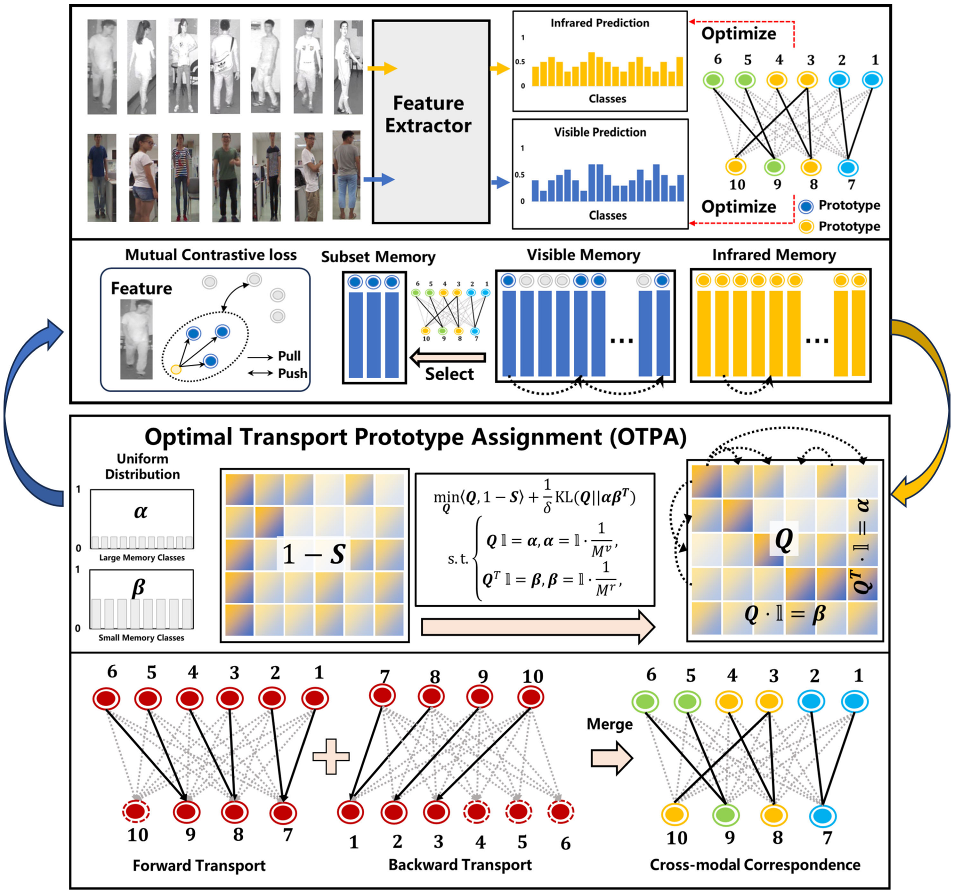

Fig. 2. The pipeline of our framework. The upper part denotes the training module, which optimizes our model using the the cross-modality correspondence from optimal transport prototype assignment (OTPA). This correspondence helps us to obtain the cross-modality pseudo labels and enables to minimize intra- and cross-modality entropy using contrastive loss, where the visible and infrared memory are established according to the clustering algorithm. In OTPA, uniform distribution and dual transporting are used to find the cross-modality correspondence. At each training epoch, two modules are alternatively processed.

Note that the first two terms of Eq. (7) encourage an average label distribution across the entire training set. The third and fourth terms of Eq. (7), *w.r.t* the minimization of entropy, suggests the model should be confident and discriminative in some samples. Besides, because of the above mentioned hypothesis with $\boldsymbol{x}^v$ and $\boldsymbol{x}^r$, *i.e.,* cross-modality correspondence, the unlabeled cross-modality data should be matched during the training process. These three prior knowledge collectively guide our proposed loop iterative training strategy. The full proof and detailed derivation are provided in Appendix A.

IV. Methodology

Suppose we are given a collection of cross-modality pedestrian images $\mathcal{X} = \{\mathcal{V}, \mathcal{R}\}$, where $\mathcal{V} = \{\boldsymbol{x}_k^v\}_{k=1}^{N^v}$ and $\mathcal{R} = \{\boldsymbol{x}_k^r\}_{k=1}^{N^r}$ denote the visible and infrared images with N^v and N^r samples, respectively. In our unsupervised setting, we do not have any real label information for training.

We follow the prior works [59] to pretrain our model. In detail, we sample a batch of visible and infrared images and forward them into a ResNet-50 based backbone $\boldsymbol{F}$ for feature extraction, *i.e.,* $\boldsymbol{f}^v = \boldsymbol{F}(\boldsymbol{x}^v)$, $\boldsymbol{f}^r = \boldsymbol{F}(\boldsymbol{x}^r)$. Then according to PCLHD [59] and PGM [22], we initialize our model by self-training.

After pre-training, we design a loop iterative training strategy alternating between model training and cross-modality matching shown in Fig. 2. During matching, optimal transport prototype assignment (OTPA) is designed to find the cross-modality correspondence among the cluster-aware prototypes. This matching process thereby generates pseudo-labels, enabling to use the prototype-based contrastive loss to minimize the intra- and cross-modality entropy. Two stages are alternatively trained, until the optimization is completed.

A. Optimal Transport Prototype Assignment (OTPA)

Different from previous work [14], we propose to find the matched visible and infrared prototypes, ensuring that the cross-modal correspondence embeds robust cross-modality correlated information while maintaining a low computational cost. To this end, we first apply DBSCAN [23] to produce clustering pseudo labels for each modality independently. With the help of cluster-aware pseudo labels, we can initialize the modality-specific prototype memories by averaging the extracted feature representations belonging to the same cluster:

$$\mathcal{M}_i^v = \frac{1}{|\boldsymbol{D}_i^v|} \sum_{\boldsymbol{f}^v \in \boldsymbol{D}_i^v} \boldsymbol{f}^v, \tag{8}$$

$$\mathcal{M}_i^r = \frac{1}{|\boldsymbol{D}_i^r|} \sum_{\boldsymbol{f}^r \in \boldsymbol{D}_i^r} \boldsymbol{f}^r, \tag{9}$$

where $\mathcal{M}_i^{v/r}$ denotes the i-th prototype in visible/infrared memory. Note that the numbers of slots in two memories are

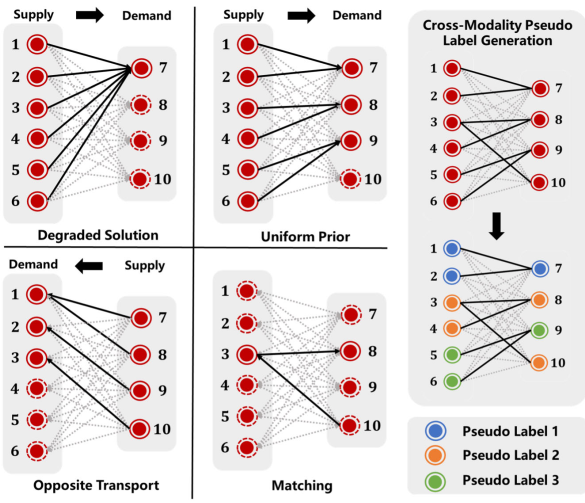

Fig. 3. Cross-modality pseudo label generation. Degraded solution: Without uniform prior, it would meet collapsed solution. Uniform prior: OTPA benefits from the balanced assignment with the uniform prior, but there are still unassigned demand prototypes, *e.g.,* node 10. Opposite transport: We conduct an opposite transport and find the matched nodes with respect to the unassigned demand prototypes, *e.g.,* node 3. Matching: final cross-modality correspondence.

different. $\boldsymbol{D}_i^{v/r}$ denotes i-th visible/infrared set from a specific cluster. $|\cdot|$ denotes the number of instances per cluster.

Thus establishing the cross-modality correspondence between each slot in $\mathcal{M}^r$ and $\mathcal{M}^v$ is actually a cross-modality matching problem. In our approach, the memory with a larger number of prototypes is viewed as supplier while the other is viewed as demander. The goal is to find a transportation plan $\boldsymbol{Q}$ that transports prototypes from suppliers to demands at the lowest transportation cost. Here the transportation cost is related to the similarities among prototypes. Thus we propose the following transport objective for solving this bipartite matching problem:

$$\min_{\boldsymbol{Q}} \left\langle \boldsymbol{Q}, 1-\frac{\boldsymbol{S}+1}{2}\right\rangle, \tag{10}$$

where $\boldsymbol{S}$ denotes the cosine similarity score matrix between the above two modality-specific prototype memories ranging from $[-1, 1]$, and $\langle\cdot\rangle$ denotes the Frobenius dot-product. Minimizing Eq. (10) implies that our transport plan favors the lower transportation cost, *i.e.,* higher cross-modality similarity.

1) Uniform Label Distribution: As suggested by mutual information guided prior, a uniform label distribution is favorable. Besides, simply optimizing Eq. (10) may lead to a collapsed solution, *i.e.,* most supply prototypes are transported to very few demand prototypes (Degraded Solution in Fig. 3). This occurs because, in the early stages of training, the model lacks discriminative capability, and therefore the similarity scores fail in accurately reflecting the cross-modal matching information.

To prevent this collapsed phenomenon, we enforce a uniform label regularization from two intuitions: 1) each supply prototype owns an equiprobable assignment choice; 2) each demand prototype owns approximately the same number of supply prototypes. To achieve intuitions, we can modify Eq. (10) by incorporating two uniformly distributed constraints (suppose visible prototypes are suppliers and infrared prototypes are demanders):

$$\min_{\boldsymbol{Q}} \left\langle \boldsymbol{Q}, 1-\frac{\boldsymbol{S}+1}{2}\right\rangle + \frac{1}{\delta}\mathrm{KL}(\boldsymbol{Q}\|\boldsymbol{\alpha}\boldsymbol{\beta}^T).$$
$$\text{s.t.} \begin{cases} \boldsymbol{Q}\mathbb{1} = \boldsymbol{\alpha}, \quad \boldsymbol{\alpha} = \mathbb{1}\cdot\dfrac{1}{|M^v|}, \\ \boldsymbol{Q}^T\mathbb{1} = \boldsymbol{\beta}, \quad \boldsymbol{\beta} = \mathbb{1}\cdot\dfrac{1}{|M^r|}, \\ \boldsymbol{S} = \mathrm{Norm}(\mathcal{M}^v)\mathrm{Norm}(\mathcal{M}^r)^T, \end{cases} \tag{11}$$

where $\boldsymbol{\alpha} \in \mathbb{R}^{M^v}$ and $\boldsymbol{\beta} \in \mathbb{R}^{M^r}$ represent the prior uniform distributed vectors, $|M^v|$ and $|M^r|$ denote the length of visible and infrared memory respectively, $\mathrm{KL}(\cdot)$ denotes the KL-divergence, $\mathrm{Norm}(\cdot)$ denotes normalized manipulation. In essence, we use KL-divergence to constrain the transportation plan $\boldsymbol{Q}$ to uniformly transport each supply prototype to the demand prototype. δ is a trade-off hyperparameter controlling the weight of this uniform regularization.

With the uniform constraint, we can adopt a fast solver of Sinkhorn-Knopp algorithm [60]. In it, the optimal solution $\hat{\boldsymbol{Q}}$ can be achieved through the iterative matrix scaling operation:

$$\forall m: \boldsymbol{\alpha}_m \leftarrow [(\boldsymbol{K})\boldsymbol{\beta}]_m^{-1} \quad \forall n: \boldsymbol{\beta}_n \leftarrow [\boldsymbol{\alpha}^T(\boldsymbol{K})]_n^{-1}, \tag{12}$$

where $\boldsymbol{K} := e^{-\delta\left(1-\frac{S+1}{2}\right)}$ is the element-wise exponential of $-\delta\left(1-\frac{S+1}{2}\right)$. Specifically, $\boldsymbol{\alpha}$ initializes with $\mathbb{1}\cdot\frac{1}{M^v}$ and $\boldsymbol{\beta}$ initializes with $\mathbb{1}\cdot\frac{1}{M^r}$. When the iteration meets the termination conditions or exceeds the maximum number, the auxiliary vectors $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are fixed. One primary advantage of this algorithm is that the optimal solution matrix of Eq. (11) can be equivalently converted as:

$$\hat{\boldsymbol{Q}} = \mathrm{diag}(\boldsymbol{\alpha})\boldsymbol{K}\mathrm{diag}(\boldsymbol{\beta}), \tag{13}$$

where $\mathrm{diag}(\cdot)$ denotes the square diagonal matrix with the elements of the corresponding vector on the main diagonal.

2) Reverse Transport: The maximal value of the optimal transport plan $\hat{\boldsymbol{Q}}$ indicates the prototype-level cross-modality correspondence, and then acted as an important semantic monitor for training the model. However, the number of visible prototypes $|M^v|$ is different from infrared ones $|M^r|$, leading to a one-to-many matching. As a result, though with uniform regularization, some demand prototypes are still not assigned.

To deal with this issue, we propose a reverse transport algorithm to merge unassigned demanders. Fig. 3 shows a toy example. If there are unassigned demander *e.g.,* node 10 in the forward transport, we would exchange the suppliers and demanders. In this case, node 10 is transported to node 3, in which this situation occurs during the reverse transport due to the many-to-one matching. Hence, node 10 and 3 are considered to be matched. Finally, we merge the nodes with cross-modality correspondence and treat them as a single class, *e.g.,* node 3, 4, 8, 10 share the same cross-modality pseudo label in this process.

After that, we construct a cross-modality prototype memory $\mathcal{M}^c$. $\mathcal{M}^c$ is initialized by averaging the cross-modality features with the same generated cross-modality pseudo labels. Note that we now have three kinds of prototype memories (*i.e.*, visible prototype memory, infrared prototype memory and cross prototype memory) and we will initialize them at the beginning of each training epoch. We use the momentum updating strategy to update the prototype memories with coefficient m at every training batch:

$$\mathcal{M}_i^v = m \cdot \mathcal{M}_i^v + (1-m) \cdot \boldsymbol{f}^v, \tag{14}$$
$$\mathcal{M}_i^r = m \cdot \mathcal{M}_i^r + (1-m) \cdot \boldsymbol{f}^r, \tag{15}$$
$$\mathcal{M}_i^c = m \cdot \mathcal{M}_i^c + (1-m) \cdot \frac{\hat{\boldsymbol{f}}^v + \hat{\boldsymbol{f}}^r}{2}, \tag{16}$$

where $\boldsymbol{f}^{v/r}$ denotes the visible or infrared feature representation in the current training batch from the i-th cluster. $\hat{\boldsymbol{f}}^v$ and $\hat{\boldsymbol{f}}^r$ are shared with the same cross-modality pseudo label. They are used to update the corresponding cross prototype. As we can see, modality-specific prototype memory $\mathcal{M}^{v/r}$ is updated by the intra-modality clustering-aware relationship, while the cross-modality prototype memory $\mathcal{M}^c$ is updated by the learned cross-modality correspondence.

Empirical results indicate the strong robustness of our transport framework. Note that previous unsupervised VI-ReID task are based on a strong assumption that the training data are exactly the same set of pedestrians in different modalities. In our approach, even when the identities from visible and infrared images are not completely overlapping, our framework remains robust and consistently improves performance.

B. Prototype-Based Contrastive Learning (PBCL)

Based on the established prototype memories, we propose prototype-based contrastive losses to minimize the intra- and cross-modality entropy.

1) Modality-Specific Contrastive Learning: Modality-specific contrastive learning consists of two parts: visible contrastive learning (VCL) and infrared contrastive learning (ICL). These two kinds of contrastive learning losses are used to minimize the intra-modality entropy and therefore promote discrimination. VCL and ICL are defined as:

$$\mathcal{L}_{VCL} = -\frac{1}{V}\sum_{k=1}^{V}\log\frac{\exp(\boldsymbol{f}_k^v \cdot \mathcal{M}_+^v/\tau)}{\sum_j \exp(\boldsymbol{f}_k^v \cdot \mathcal{M}_j^v/\tau)}, \tag{17}$$
$$\mathcal{L}_{ICL} = -\frac{1}{R}\sum_{k=1}^{R}\log\frac{\exp(\boldsymbol{f}_k^r \cdot \mathcal{M}_+^r/\tau)}{\sum_j \exp(\boldsymbol{f}_k^r \cdot \mathcal{M}_j^r/\tau)}, \tag{18}$$

where V/R denotes the number of visible/infrared samples in each training batch, $\mathcal{M}_+^{v/r}$ denotes the positive modality-specific prototype corresponding to the clustering-aware pseudo label of the k-th query feature $\boldsymbol{f}_k$, and τ denotes the temperature hyper-parameter.

2) Mutual Contrastive Learning (MCL): To further minimize the cross-modality entropy, we dig up the correlations between the above two kinds of pseudo labeling frameworks, *i.e.,* intra-modality clustering-aware pseudo labels and cross-modality pseudo labels generated from OTPA, so as to construct prototype-based contrastive learning loss. In this loss, for each sample in the training batch, we first compute the cosine similarity between its feature representation and the modality-specific prototypes of the opposite modality. Taking the visible query feature as an example, we can find the cosine similarity as:

$$s_{ki}^v = \text{Norm}(\boldsymbol{f}_k^v)^T \text{Norm}(\hat{\mathcal{M}}_i^r), \tag{19}$$

where $\hat{\mathcal{M}}^r$ denotes a subset of $\mathcal{M}^r$ and Norm$(\cdot)$ denotes normalized manipulation. To construct $\hat{\mathcal{M}}^r$, we select the prototypes in $\mathcal{M}^r$ sharing the same cross-modality pseudo label when given a query visible sample. Note that the prototypes in $\hat{\mathcal{M}}^r$ may come from one or several kinds of DBSCAN clusters. Then if this cosine similarity score exceeds a threshold ($\theta = 0.2$), we treat them as the positive similarity scores $\hat{s}_{ki}^v$. Thus, the loss function can be formulated as:

$$\mathcal{L}_{MCL} = -\frac{1}{V}\sum_{k=1}^{V}\log\frac{\sum_i \exp(\hat{s}_{ki}^v/\tau)}{\sum_j \exp(\boldsymbol{f}_k^v \cdot \mathcal{M}_j^r/\tau)} - \frac{1}{R}\sum_{k=1}^{R}\log\frac{\sum_i \exp(\hat{s}_{ki}^r/\tau)}{\sum_j \exp(\boldsymbol{f}_k^r \cdot \mathcal{M}_j^v/\tau)}. \tag{20}$$

C. Optimization

In our loop training, we fine-tuning the model using the complete loss functions below based on the pretraining weights:

$$\mathcal{L} = \mathcal{L}_D + \lambda\mathcal{L}_{Tri} + \mathcal{L}_{VCL} + \mathcal{L}_{ICL} + \mathcal{L}_{MCL} + \mathcal{L}_{CPAL}, \tag{21}$$

where $\mathcal{L}_{Tri}$ denotes the intra-modality triplet loss by giving cross-modality pseudo labels generated by OTPA, $\mathcal{L}_{CPAL}$ denotes the cross prediction alignment learning loss (CPAL) from OTLA-ReID [14] (the prediction here is defined as the cosine similarities between the query features and the prototypes in the cross prototype memory, where more details will be shown in Appendix B.1). $\mathcal{L}_D$ represents the Modality Discriminator Loss adopted from OTLA-ReID [14], which employs a Discrepancy Elimination Network (DEN) to reduce the modality gap via adversarial training. The $\lambda = 0.5$ is a hyper-parameter. Other weights of loss functions are simply set to 1.0.

V. Experiment

A. Experimental Settings

1) Datasets Summary: The proposed method has been evaluated on two widely adopted benchmarks **SYSU-MM01** [73] and **RegDB** [74]. SYSU-MM01 is a large-scale cross-modality ReID dataset that is collected by four RGB and two near-infrared cameras from both indoor and outdoor environments. It's composed of 287,628 visible images and 15,792 infrared images in total of 491 identities. RegDB is collected by two aligned cameras (one visible and one far-infrared). It includes 412 identities and each identity has 10 infrared images and 10 visible images. Hence RegDB totally contains 4,120 visible images and 4,120 far-infrared images.

TABLE I

COMPARISONS WITH SOTA METHODS ON SYSU-MM01 OF ALL SEARCH MODE AND INDOOR SEARCH MODE, INCLUDING UNSUPERVISED REID (USL-REID), FULLY-SUPERVISED VI-REID (SVI-REID) AND UNSUPERVISED VI-REID (USVI-REID). ALL METHODS ARE MEASURED BY CMC(%), MAP(%) AND MINP(%). † INDICATES WE RE-IMPLEMENT THE RESULT WITH THE OFFICIAL CODE. BOLD INDICATES THE BEST PERFORMANCE AMONG UNSUPERVISED METHODS, AND UNDERLINED INDICATES THE BEST PERFORMANCE AMONG ALL METHODS

Settings			All Search					Indoor Search				
Type	Method	Venue	Rank-1	Rank-10	Rank-20	mAP	mINP	Rank-1	Rank-10	Rank-20	mAP	mINP
USL-ReID	MetaCam† [75]	CVPR'21	14.7	32.7	40.5	9.3	-	23.9	43.8	51.6	17.1	-
	HCD† [24]	ICCV'21	18.0	48.8	60.9	17.9	-	24.4	61.9	76.1	28.8	-
	ICE [76]	ICCV'21	20.5	57.5	70.9	10.2	20.4	29.8	69.4	82.7	38.4	34.3
	STS† [5]	TIP'22	18.7	51.8	63.9	19.9	-	26.8	77.5	31.3	-	
	PPLR† [4]	CVPR'22	19.1	54.1	66.4	18.6	-	20.6	58.5	71.9	26.3	-
	DCMIP† [25]	ICCV'23	21.3	58.7	71.9	19.5	-	25.5	65.9	84.2	40.2	-
SVI-ReID	AGW [6]	TPAMI'21	47.5	84.4	92.1	47.7	35.3	54.2	91.1	96.0	63.0	59.2
	MPANet [77]	CVPR'21	70.6	96.2	98.8	68.2	-	76.7	98.2	99.6	81.0	-
	SPOT [32]	TIP'22	65.3	92.7	97.0	62.3	48.9	69.4	96.2	99.1	74.6	70.5
	MSCLNet [41]	ECCV'22	77.0	97.6	<u>99.2</u>	71.6	-	78.5	<u>99.3</u>	<u>99.9</u>	81.2	-
	SEFL [42]	CVPR'23	77.1	97.0	99.1	72.3	-	82.1	97.4	98.9	83.0	-
	PartMix [34]	CVPR'23	<u>77.8</u>	-	-	74.6	-	81.5	-	-	84.4	-
	RLE [43]	NIPS'24	75.4	<u>97.7</u>	-	72.4	-	84.7	<u>99.3</u>	-	87.0	-
	WRIM-Net [36]	ECCV'24	77.4	-	-	<u>75.4</u>	-	<u>86.2</u>	-	-	<u>88.1</u>	-
	TSKD [78]	PR'25	76.6	97.1	99.1	73.0	-	82.7	98.9	99.8	85.3	-
USVI-ReID	H2H [16]	TIP'21	25.5	63.9	76.2	25.2	-	13.9	30.4	39.3	12.7	-
	OTLA-ReID [14]	ECCV'22	29.9	71.8	83.9	27.1	-	29.8	74.1	87.6	38.8	-
	ADCA [17]	ACM MM'22	45.5	85.3	93.2	42.7	28.3	50.6	89.7	96.2	59.1	55.2
	CHCR [18]	TCSVT'23	47.7	87.3	94.2	45.3	-	50.1	90.8	95.6	42.2	-
	DOTLA [19]	ACM MM'23	50.4	89.0	95.9	47.4	32.4	53.5	92.2	97.8	61.7	57.4
	CCLNet [21]	ACM MM'23	54.0	88.8	95.0	50.2	-	56.7	91.1	97.2	65.1	-
	PGM [22]	CVPR'23	57.3	92.5	97.2	51.8	35.0	56.2	90.2	95.4	62.7	58.1
	MMM [55]	ECCV'24	61.6	-	-	57.9	-	64.4	-	-	70.4	-
	PCLHD [59]	NIPS'24	64.4	-	-	58.7	-	69.5	-	-	74.4	-
	RoDE [57]	TIFS'25	62.9	-	-	57.9	43.0	64.5	-	-	70.4	66.0
	DLM [56]	TPAMI'25	62.2	**93.4**	97.4	58.4	43.7	67.3	**95.9**	**99.2**	72.9	68.9
	OTPA-ReID	-	**69.4**	**93.4**	**97.5**	**63.2**	**47.7**	**74.9**	94.8	98.1	**77.4**	**<u>73.3</u>**

2) Evaluation Metrics: On both datasets, we follow the popular protocols [13] for evaluation, in which cumulative match characteristic (CMC), mean average precision (mAP), and mean inverse negative penalty (mINP) [6] are adopted. SYSU-MM01 contains two different testing settings, *i.e., All Search* mode and *Indoor Search* mode. For all-search mode, the gallery set consists of all visible images (captured by CAM1, CAM2, CAM4, CAM5), and the query set is composed of all infrared samples (captured by CAM3, CAM6). For indoor-search mode, images are captured only from the indoor scene to constitute the gallery set, excluding the images from CAM4 and CAM5. On both search modes, the proposed method is evaluated under the single-shot setting. For RegDB, we randomly select 206 identities for training and the remaining 206 identities for testing. It also includes two different settings, *i.e.*, *Visible to Thermal* mode and *Thermal to Visible* mode. We report the average result by randomly splitting of training and testing set over 10 times.

B. Implementation Details

1) Training Details: We implement our method using PyTorch on one NVIDIA TITAN RTX GPU. As a preliminary step, we replicate the experimental setup of PCLHD [59] by adhering to its original configuration. Following this, we fine-tune the trained model parameters using the proposed OTPA method for 20 epochs. The batch size is fixed to 64 for all experiments. According to the cross-modality pseudo labels from OTPA, in every batch, we sample 4 different identities (pseudo labels), and each identity includes 8 visible images and 8 infrared images. The model is optimized by Adam optimizer [79] with an initial learning rate of 3.5×10^{-3}. All the pedestrian images are resized to 288×144. The margin ρ of triplet loss is set to 0.3, and the δ of OTPA is fixed at 25. The temperature τ is equal to 0.05. Besides, the input images are randomly flipped and erased with 50% probability, while visible images are extra randomly augmented by grayscale with 50% probability. The momentum coefficient m in memory update is set to 0.1.

We adopt ResNet-50 [80] pretrained on ImageNet [81] as our backbone. The first shallow block is modality-independent and the remaining blocks are shared. The modality classifier in DEN is implemented with three FC layers and one BN layer [82].

C. Main Results

To demonstrate the effectiveness of our method, we compare our approach with four related ReID settings, including fully-supervised VI-ReID (SVI-ReID), unsupervised VI-ReID (USVI-ReID), and unsupervised learning ReID (USL-ReID). As for USL-ReID, it's designed for the visible modality unsupervised task. Here, we use both unlabeled visible and infrared data to re-implement methods of this setting, and therefore those methods would naturally perform poorly compared with other settings. The main results are shown in Tab. I and II.

1) Comparison With Unsupervised Methods: As shown in the last block of Tab. I and II, here are the key observations: 1)

TABLE II

COMPARISONS WITH SOTA METHODS ON REGDB OF VISIBLE TO THERMAL MODE AND THERMAL TO VISIBLE MODE, INCLUDING UNSUPERVISED REID (USL-REID), FULLY-SUPERVISED VI-REID (SVI-REID) AND UNSUPERVISED VI-REID (USVI-REID). ALL METHODS ARE MEASURED BY CMC(%), MAP(%) AND MINP(%). BOLD INDICATES THE BEST PERFORMANCE AMONG UNSUPERVISED METHODS, AND UNDERLINED INDICATES THE BEST PERFORMANCE AMONG ALL METHODS

Settings			Visible to Thermal					Thermal to Visible				
Type	Method	Venue	Rank-1	Rank-10	Rank-20	mAP	mINP	Rank-1	Rank-10	Rank-20	mAP	mINP
USL-ReID	MetaCam† [75]	CVPR'21	23.1	44.6	55.3	17.5	-	20.9	43.3	50.8	16.5	-
	HCD† [24]	ICCV'21	10.8	20.6	27.2	12.3	-	12.4	26.5	33.9	13.7	-
	ICE [76]	ICCV'21	13.0	25.9	34.4	15.6	11.9	12.2	25.7	34.9	14.8	10.6
	STS † [5]	TIP'22	11.7	25.8	32.8	14.9	-	12.5	25.1	32.5	13.3	-
	PPLR† [4]	CVPR'22	12.4	26.2	34.2	13.2	-	8.6	20.7	28.8	10.7	-
	DCMIP† [25]	ICCV'23	25.1	45.8	57.2	19.0	-	24.9	44.9	56.2	18.2	-
SVI-ReID	AGW [6]	TPAMI'21	70.1	86.2	91.6	66.4	50.2	70.5	87.2	91.8	65.9	51.2
	MPANet [77]	CVPR'21	83.7	-	-	80.9	-	82.8	-	-	80.7	-
	SPOT [32]	TIP'22	80.4	93.5	96.4	72.5	56.2	79.4	92.8	96.0	72.3	56.1
	MSCLNet [41]	ECCV'22	84.2	-	-	81.0	-	83.9	-	-	78.3	-
	SEFL [42]	CVPR'23	70.2	86.1	-	52.5	-	67.7	84.7	-	52.3	-
	PartMix [34]	CVPR'23	85.7	-	-	82.3	-	84.9	-	-	82.5	-
	RLE [43]	NIPS'24	92.8	**97.9**	-	88.6	-	91.0	<u>97.5</u>	-	86.6	-
	WRIM-Net [36]	ECCV'24	<u>94.5</u>	-	-	**90.5**	-	<u>93.7</u>	-	-	<u>89.7</u>	-
	TSKD [78]	PR'25	91.1	-	-	81.7	-	89.9	-	-	80.5	-
USVI-ReID	H2H [16]	TIP'21	23.8	45.3	54.0	18.9	-	-	-	-	-	-
	OTLA-ReID [14]	ECCV'22	32.9	54.2	62.5	29.7	-	32.1	56.1	64.9	28.6	-
	ADCA [17]	ACM MM'22	67.2	82.0	87.4	64.1	52.7	68.5	83.2	88.0	63.8	49.6
	CHCR [18]	TCSVT'23	68.2	81.5	88.7	63.8	-	69.1	83.7	88.2	64.0	-
	DOTLA [19]	ACM MM'23	85.6	94.1	95.5	76.7	61.6	82.9	92.3	94.9	75.0	58.6
	CCLNet [21]	ACM MM'23	69.9	-	-	65.5	-	70.2	-	-	66.7	-
	PGM [22]	CVPR'23	69.5	-	-	65.4	-	69.9	-	-	65.2	-
	MMM [55]	ECCV'24	89.7	-	-	80.5	-	85.8	-	-	77.0	-
	PCLHD [59]	NIPS'24	84.3	-	-	80.7	-	82.7	-	-	78.4	-
	RoDE [57]	TIFS'25	88.8	-	-	79.0	68.0	85.8	-	-	78.4	62.3
	DLM [56]	TPAMI'25	87.6	94.7	96.7	82.8	<u>**71.9**</u>	86.8	94.4	96.4	81.9	<u>**69.0**</u>
	OTPA-ReID	-	**89.4**	**95.6**	<u>**97.0**</u>	**83.0**	70.6	**88.2**	**95.1**	<u>**96.6**</u>	**82.1**	68.3

To evaluate the effectiveness of the proposed OTPA method, we compare it with ten state-of-the-art unsupervised VI-ReID methods. The results indicate that our method significantly outperforms existing methods.PCLHD [59] is a progressive contrastive learning method designed for better cross-modality feature learning. By applying our OTPA method to fine-tune the PCLHD model, we achieve a remarkable improvement, raising Rank-1 accuracy to 69.4% and mAP to 63.2%. This demonstrates that OTPA enhances the original PCLHD framework by optimizing the cross-modality assignment, leading to superior feature alignment and retrieval performance. 2) In addition, some unsupervised VI-ReID methods like H2H [16] and OTLA-ReID [14] need an extra labeled visible dataset for auxiliary training. Compared with them, our method achieves significant gains on both datasets. 4) For other unsupervised methods designed for single-modality ReID tasks, our experimental results surpass them by a significant margin. It is somewhat unfair to directly compare them since most of them don't consider the cross-modality discrepancy. We report here to highlight the difference between the unsupervised VI-ReID task and the unsupervised ReID task.

2) Comparison With Fully-Supervised Methods: We provide some currently well-known fully-supervised VI-ReID results for reference. Surprisingly, our method outperforms a part of fully-supervised VI-ReID methods on both SYSU-MM01 and RegDB datasets (especially on RegDB), including SEFL [42], Partmix [34], RLE [43] WRIM-Net [36] and TSKD [78]. Such phenomenon indicates that label information of both modalities could be gradually aligned by our proposed optimal-transport assignment algorithm. Thus, the unsupervised training process can be gradually closed to the supervised one. Besides, we should admit that there is still some gap between our method and SOTA fully-supervised results.

D. Ablation Study

In this subsection, we conduct the ablation study to show the effectiveness of each component in our approach. Concrete experimental results are shown in Tab. III and Tab. IV. We have pretrained our model before our proposed training stage, which has drawn inspiration from the strong baseline of current USVI-ReID methods, *e.g.*, PCLHD [59]. The first row in the tables represents the "Baseline" result derived from this pre-trained model. Without OTPA algorithm, we can only use the cluster-aware pseudo labels to sample training data and compute $\mathcal{L}_{Tri}$. After adopting optimal transport prototype assignment strategy (denoted as "+ OTPA"), we use generated cross-modality pseudo labels to sample training data and compute $\mathcal{L}_{Tri}$. Besides, we can present a complete loss framework by adding $\mathcal{L}_{CPAL}$, $\mathcal{L}_{MCL}$, and $\mathcal{L}_{D}$ incrementally. The last row represents the "Full Model" with all components enabled.

1) Effectiveness of OTPA Algorithm: Apparently, benefiting from OTPA, we achieve a significant promotion of final performance on both datasets (around 28% Rank-1 (10% Rank-1) and 25% mAP (10% mAP) on SYSU-MM01 (RegDB), by comparing the "Baseline" and the "Full Model" in Tab. III (Tab. IV). It appears that OTPA allows us to dig up the

TABLE III

ABLATION STUDY IN TERMS OF CMC(%), MAP(%), AND MINP(%) ON SYSU-MM01. THE BASELINE INCLUDES $\mathcal{L}_{VCL}$, $\mathcal{L}_{ICL}$, AND $\mathcal{L}_{Tri}$. EACH ROW REPRESENTS THE RESULT OF ADDING A SPECIFIC COMPONENT TO THE CONFIGURATION OF THE PREVIOUS ROW

Method Configuration	All Search			Indoor Search		
	Rank-1	mAP	mINP	Rank-1	mAP	mINP
Baseline	41.65	38.76	24.55	47.51	56.03	51.72
+ OTPA	61.06	58.46	43.42	70.61	73.91	69.26
+ $\mathcal{L}_{CPAL}$	62.56	58.58	43.11	71.11	74.03	69.04
+ $\mathcal{L}_{MCL}$	67.97	62.23	46.50	73.46	76.60	72.48
+ $\mathcal{L}_{D}$ **(Full Model)**	**69.37**	**63.16**	**47.74**	**74.91**	**77.42**	**73.25**

TABLE IV

ABLATION STUDY IN TERMS OF CMC(%), MAP(%), AND MINP(%) ON REGDB. THE BASELINE INCLUDES $\mathcal{L}_{VCL}$, $\mathcal{L}_{ICL}$, AND $\mathcal{L}_{Tri}$. EACH ROW REPRESENTS THE RESULT OF ADDING A SPECIFIC COMPONENT TO THE CONFIGURATION OF THE PREVIOUS ROW

Method Configuration	Visible to Thermal			Thermal to Visible		
	Rank-1	mAP	mINP	Rank-1	mAP	mINP
Baseline	80.37	72.54	56.45	79.13	70.96	54.79
+ OTPA	83.56	77.31	63.17	82.88	75.97	59.78
+ $\mathcal{L}_{CPAL}$	87.67	81.08	67.70	86.20	79.91	65.51
+ $\mathcal{L}_{MCL}$	87.63	81.33	68.50	86.39	80.33	66.36
+ $\mathcal{L}_{D}$ **(Full Model)**	**89.35**	**82.99**	**70.58**	**88.22**	**82.09**	**68.31**

cross-modality correspondence and thereby enables us to mine the extra cross-modality supervised signals.

2) Effectiveness of Prototype-Based Contrastive Learning: Here, we analyze the effectiveness of prototype-based contrastive learning losses. First of all, benefiting from the strong baseline PCLHD [59] and the assistance of $\mathcal{L}_{VCL}$ and $\mathcal{L}_{ICL}$, the "+ OTPA" baseline both achieve relatively good results, which are superior to the overall performance of OTLA-ReID as shown in Tab. I and Tab. II. We can also find that cross prediction alignment learning can boost performance when combined with other components (see "+$\mathcal{L}_{CPAL}$" in Tab. III and Tab. IV, especially on RegDB). That indicates a further promotion would be expected when aligning the batch-level prototype-based predictions with the mix-up technique between two modalities. Besides, the mutual contrastive learning ($\mathcal{L}_{MCL}$) encourages exploring the complementary knowledge between the clustering pseudo labels and cross-modality pseudo labels, which can achieve great improvements (see "+$\mathcal{L}_{MCL}$" in Tab. III and Tab. IV) as well.

E. Discussion

To isolate the impact of strong baselines and facilitate a more rigorous discussion of our proposed OTPA's efficacy, we specifically devised and implemented a training framework, termed OTPA-ReID-FS. We integrate the pretraining stage (commonly used in ADCA [17] and PGM [22]) with our proposed training stage, training our model end-to-end from scratch to ensure unbiased evaluation. Training details are provided in supplementary materials.

1) Training Time Analysis: As illustrated in the Fig. 5(a), we show the total training time of OTLA-ReID, OTLA running time (the key component in OTLA-ReID [14] which are most related to OTPA), total training time of our proposed OTPA-ReID-FS, DBSCAN running time, and OTPA running time every 10 training epochs. Note that we pretrain the model of OTPA-ReID-FS with 40 (20) epochs on SYSU-MM01 (RegDB) so the OTPA running time of the pretraining epoch is zero. Apparently, we can see OTPA algorithm is very efficient for both datasets as it only transports the prototypes that are relatively fewer in number, where the running time is negligible (around 0.01s for SYSU-MM01 and 0.005s for RegDB) compared with OTLA algorithm. However, OTPA algorithm introduces the extra DBSCAN for preliminary clustering (around the 150s for SYSU-MM01 and 15s for RegDB) due to the lack of annotation. Actually, OTLA algorithm also necessitates an additional UDA process, which is more time-consuming and resource-intensive, to generate reliable visible pseudo labels in the unsupervised setting.

2) Pseudo Labels Matching Accuracy: To illustrate the accuracy of our generated pseudo labels, we show the matching accuracy of cross-modality pseudo labels generated by OTPA and OTLA (unsupervised) in Fig. 5(b). This epoch-level matching accuracy is defined by the average proportion of our learned cross-modal correspondence and the ground-truth for the cross-modality data pairs. The cross-modality data pairs are constructed by randomly selecting two samples sharing the same cross-modality pseudo label (Note that both OTLA and OTPA can produce the matching data pair). Due to the initial 40 (20) training epochs of OTPA-ReID-FS dedicated to pretraining on SYSU-MM01 (RegDB), only cluster-aware pseudo labels are available for sampling training data. Therefore, the corresponding matching accuracy is nearly zero. The visualization results show that the pseudo label matching accuracy of OTLA-ReID and OTPA-ReID-FS iteratively improved as the training goes on. However, OTPA-ReID-FS can finally reach around 58% on SYSU-MM01 and around 95% on RegDB significantly outperforming OTLA-ReID. This phenomenon indicates that the OTPA algorithm can more effectively explore the correct cross-modality correspondence. Besides, this pseudo label matching accuracy is very close to the Rank-1 accuracy, which can be an important indicator for the USVI-ReID task.

3) Visualization of Feature Embedding Space: We visualize the feature embedding space with t-SNE [83] by projecting the features into a 2D plane. We randomly select the cross-modality samples from 20 classes in Fig. 6. It appears that

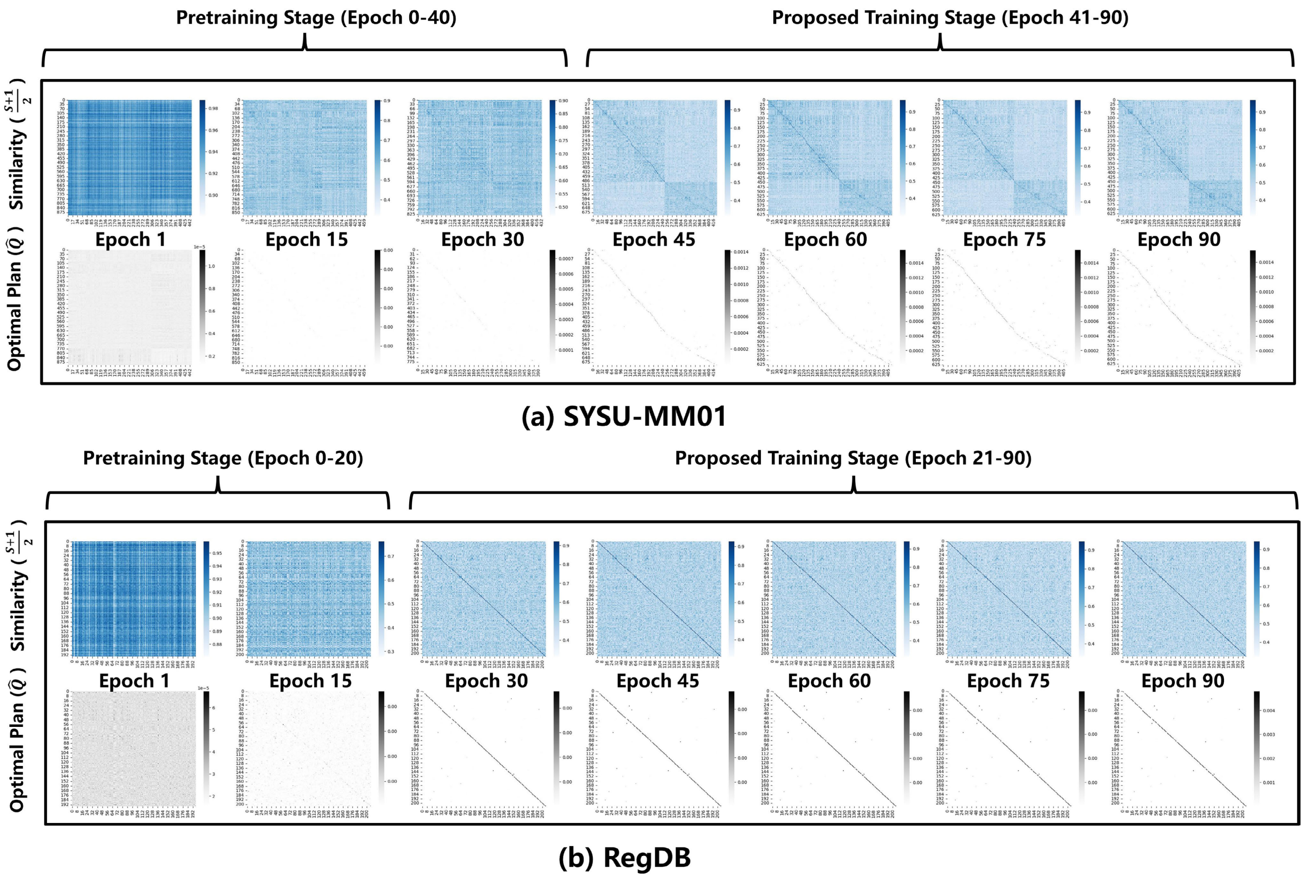

Fig. 4. Visualization of optimal transport prototype assignment (OTPA) algorithm. We visualize the normalized cosine similarity matrix $\frac{S+1}{2}$ (input of OTPA algorithm ranged from [0, 1]) and the optimal transport plan $\hat{Q}$ (output of OTPA algorithm). The horizontal and vertical coordinates of each visualized matrix denote visible pseudo classes and infrared pseudo classes, respectively.

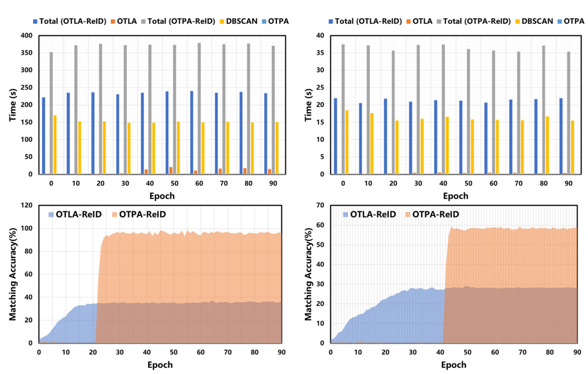

Fig. 5. (a) Quantitative analysis of training time. We show the total training time of OTLA-ReID, OTLA running time (the key component in OTLA-ReID [14] which are most related to OTPA), total training time of our proposed OTPA-ReID-FS, DBSCAN running time, and OTPA running time every 10 training epochs. (b) The comparison of pseudo label matching accuracy for OTLA-ReID [14] and OTPA-ReID-FS in each training epoch.

at the pretraining stage (*i.e.*, epoch 0-40 for SYSU-MM01 and epoch 0-20 for RegDB), the inter-modality discrepancy is large so that there are obvious intervals between the cross-modality sample points. We can also find that sample points of different classes within one modality can not be separated explicitly in this initial stage. However, when incorporating our proposed method, the points with the same ground-truth labels are gradually gathering together as the training epoch increases no matter what modality they come from. This phenomenon vividly validates the effectiveness of our proposed method, although there still exists wrongly assembled sample points.

4) *Visualization of OTPA Algorithm:* As shown in Fig. 4(a) and Fig. 4(b), we visualize the optimal transport results. The upper parts of this figure depict the normalized cosine similarity matrices $\frac{S+1}{2}$ (input of OTPA algorithm ranged from $[0, 1]$) computed between the prototypes from two modality-specific memories, while the bottom parts show the optimal transport plan $\hat{Q}$ (output of OTPA algorithm). Darker colors indicate higher values. The horizontal and vertical coordinates of each visualized matrix denote the pseudo classes for visible and infrared prototype memory, respectively. Note that we do not use OTPA algorithm for pretraining so we implement it here only for better visualization and comparison.

As observed, irrespective of the utilization of the OTPA algorithm, we are unable to establish clear cross-modality correspondence during the pretraining stage. This is because the pretraining sampling strategy we used relies on intra-modality clustering results. It also can be seen that as our proposed training stage progresses, the cross-modality correspondence becomes more and more explicit, as it is optimized by our designed OTPA algorithm and contrastive learning losses. Furthermore, it seems that OTPA algorithm can generate more

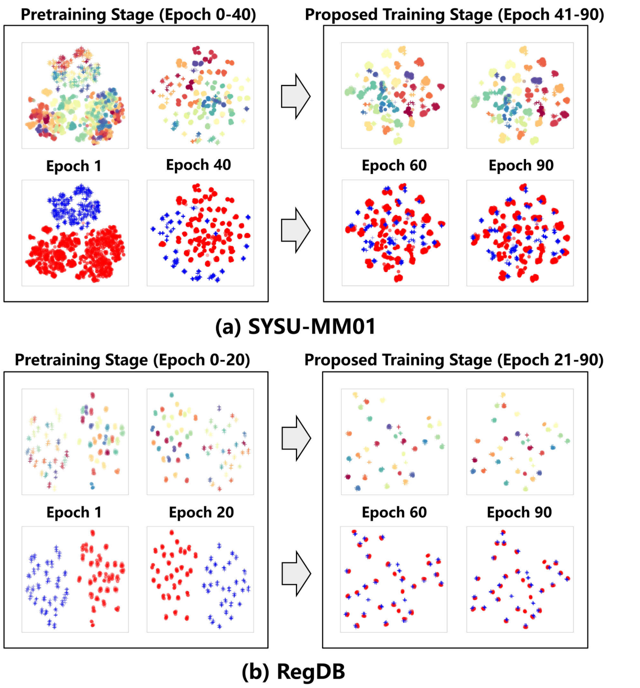

Fig. 6. Visualization of feature embedding space with t-SNE [83] on SYSU-MM01 and RegDB, where we randomly select cross-modality samples from 20 classes. The circle symbol (∘) denotes the instance from visible modality and the cross symbol (+) denotes the instance from infrared modality. The different colors of the upper figure of each dataset indicate the samples from different classes, while in the bottom figure, two kinds of colors only reflect the modality information.

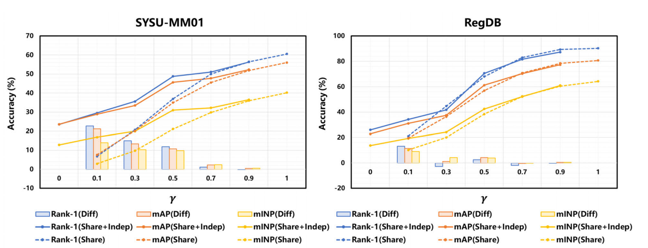

Fig. 7. Performance under the Share mode and the Share+Indep mode as the γ value changing. We also visualize the performance's difference (Diff, *i.e.*, Share+Indep mode minus Share mode) of the same γ value.

explicit correspondence compared with the cosine similarity matrix (*i.e.,* without OTPA algorithm). Besides, with the uniform prior (*i.e.,*$\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$) in OTPA, the degeneration phenomenon (*i.e.*, most prototypes of one modality are assigned to few prototypes from another modality) will be relieved in the transporting process.

5) Analysis of Inconsistent Class Space in Training Data: As we can see, the majority of recent works [14], [16], [17], [18], [19], [20], [21], [22] on the unsupervised VI-ReID task (including ours) are based on a strong assumption that the training data are exactly the same set of pedestrians in different modalities. To explore the influence of this assumption, we have done extra experiments (*i.e.*, we select a subset of the original training set, and then set a hyper-parameter γ to control the proportion of the overlapped classes between the visible subset and the infrared subset. The remaining classes are randomly divided with equal proportion for each modality).

For example, suppose there are 4 categories (*i.e.*, 1, 2, 3, 4) in the training set. If we set $\gamma = 0.5$, then we randomly select 2, 3 as the overlapped classes. Share mode indicates we only take the cross-modality samples from 2, 3 for training. Share+Indep mode indicates that the visible training set contains categories 1, 2, 3 while the infrared set contains categories 2, 3, 4. For better comparison, we also do the experiments of Share mode by only using the category overlapped cross-modality training data. Specifically, we report the average experimental results in Tab. V with γ ranging from 0 to 1. Note that each mode here is conducted by repeating the experiment 3 times to eliminate the effect of the random seed. Besides, we also visualize the model's performance under the Share mode and the Share+Indep mode as the γ value changing in Fig. 7. We also add the performance's difference (Diff, *i.e.*, Share+Indep mode minus Share mode) of the same γ value into the Fig. 7.

From Tab. V and Fig. 7, we can find that our proposed method is robust and consistently improves the performances compared with the pretraining stage (note that the pretraining and our cross-modality training use the same data), even the the classes of each modality are incompletely non-overlapping. When γ is below than 0.9, Share+Indep mode exceeds the Share mode. When the classes are nearly overlapping (γ = 0.9), it appears that these two modes achieve comparable performance.

In conclusion, we can see this assumption is necessary and important in the unsupervised cross-modality person ReID task. Surprisingly, even when the classes of each modality are incompletely overlapping, our approach still works. This is because identity-aware discrimination does not rely solely on fully supervised signals. For example, people dressed similarly also can provide useful information. On the other hand, it is usual to collect data from a supermarket or mall during a period of time. Thus one person appearing in multiple cameras is quite common.

6) Limitations: While our approach demonstrates robust performance in unsupervised visible-infrared person re-identification, several inherent limitations warrant discussion. Primarily, the reliance on pseudo-labeling introduces inevitable label noise; although our Optimal Transport Prototype Assignment (OTPA) mitigates this via global matching constraints, extreme cross-modality discrepancies or semantic gaps can still result in mismatches that compromise the purity of prototype memory. This issue exacerbates the risk of overfitting to false correlations, particularly in early training epochs or on smaller datasets, despite the regularization provided by contrastive losses. Furthermore, our optimal transport formulation is predicated on the assumption that a balanced cross-modal correspondence is achievable. In edge scenarios characterized by severe occlusion or thermal crossover where modal similarity is negligible, this assumption may be challenged, leading to potential alignment failures. To address these complexities, our future work will focus on integrating uncertainty estimation to filter unreliable pseudo-labels and exploring more flexible matching mechanisms to enhance robustness in extreme scenarios.

TABLE V

THE PERFORMANCE COMPARISON BETWEEN THE PERTAINING STAGE (COMMONLY USED IN ADCA [17] AND PGM [22]) AND OUR PROPOSED TRAINING STAGE WITH DIFFERENT γ VALUES ON SYSU-MM01 (ALL SEARCH) AND REGDB (VISIBLE TO THERMAL). γ IS USED TO CONTROL THE PROPORTION OF THE OVERLAPPING CLASSES BETWEEN THE VISIBLE SUBSET AND THE INFRARED SUBSET. WE PROPOSE THE SHARE MODE AND SHARE+INDEP MODE TO VERIFY THE ROBUSTNESS OF OUR METHOD. ALL EXPERIMENTS ARE MEASURED BY CMC(%), MAP(%) AND MINP(%)

Settings			SYSU-MM01 (All Search)			RegDB (Visible to Thermal)		
Mode	γ value	Stage	Rank-1	mAP	mINP	Rank-1	mAP	mINP
Share + Indep	0.0	Pretrain	24.50	24.59	13.90	10.53	12.20	7.97
		Train	23.56 (-0.94)	23.59 (-1.00)	12.83 (-1.07)	26.02 (+15.49)	22.83 (+10.63)	13.70 (+5.73)
	0.1	Pretrain	26.93	26.47	14.70	12.23	12.88	8.15
		Train	29.57 (+2.64)	28.90 (+2.53)	16.86 (+2.16)	34.27 (+22.04)	31.06 (+18.18)	19.31 (+11.16)
	0.3	Pretrain	29.39	28.36	15.82	14.61	15.21	8.44
		Train	35.63 (+6.24)	33.50 (+5.14)	19.94 (+4.12)	41.84 (+27.23)	37.53 (+22.32)	24.24 (+15.79)
	0.5	Pretrain	34.98	33.81	20.71	20.29	20.21	12.13
		Train	48.80 (+13.82)	45.64 (+11.83)	31.01 (+10.3)	70.58 (+50.29)	61.25 (+41.04)	42.48 (+30.35)
	0.7	Pretrain	34.16	32.85	19.68	23.45	21.91	13.25
		Train	51.11 (+16.95)	47.75 (+14.90)	32.22 (+12.54)	81.70 (+58.25)	70.39 (+48.48)	52.21 (+38.96)
	0.9	Pretrain	34.77	33.75	21.06	26.45	25.69	17.45
		Train	56.35 (+21.58)	52.32 (+18.57)	36.53 (+15.47)	87.33 (+60.88)	77.41 (+51.72)	61.02 (+43.57)
Share	0.1	Pretrain	5.37	6.97	2.84	4.37	5.67	3.66
		Train	6.81 (+1.44)	7.62 (+0.65)	2.92 (+0.08)	21.17 (+16.8)	19.57 (+13.9)	10.20 (+6.54)
	0.3	Pretrain	15.23	16.00	7.74	6.99	7.39	4.52
		Train	20.71 (+5.48)	20.16 (+4.16)	9.72 (+1.98)	44.71 (+37.72)	36.41 (+29.02)	20.05 (+15.53)
	0.5	Pretrain	23.04	23.03	12.47	5.83	8.38	5.07
		Train	36.97 (+13.93)	34.96 (+11.93)	21.17 (+8.7)	68.06 (+62.23)	57.02 (+48.64)	38.46 (+33.39)
	0.7	Pretrain	32.04	30.70	17.89	15.49	15.90	9.76
		Train	49.99 (+17.95)	45.53 (+14.83)	29.84 (+11.95)	83.06 (+67.57)	70.86 (+54.96)	52.40 (+42.64)
	0.9	Pretrain	33.46	32.38	19.29	21.12	21.27	12.75
		Train	56.47 (+23.01)	51.91 (+19.53)	36.03 (+16.74)	89.37 (+68.25)	78.47 (+57.23)	60.48 (+48.03)
	1.0	Pretrain	36.30	35.41	22.14	27.65	26.45	18.54
		Train	60.58 (+24.28)	56.07 (+20.66)	40.25 (+18.11)	90.34 (+62.69)	80.72 (+54.27)	64.29 (+45.75)

VI. CONCLUSION

In this paper, we propose a mutual information guided optimal transport approach for unsupervised VI-ReID. In this framework, three learning principles, *i.e.*, entropy minimization, uniform label distribution and cross-modality matching are derived based on the proposed theoretical objective. To reach these principles, a loop iterative training strategy alternating between model training and cross-modality matching is designed. In the matching stage, we formulate the cross-modality prototype matching problem as an optimal transport task, which allows for finding the cross-modality correspondence. In the training stage, prototype-based contrastive learning losses are designed for cross-modality learning based on this correspondence. In future work, we would like to do more theoretical analysis and extend this framework to broader multi-modal scenarios. For instance, in unsupervised vision-language learning, our method could be adapted to align visual regions with textual phrases by treating them as cross-modal prototypes, thereby facilitating fine-grained semantic alignment without relying on massive paired annotations.

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